Magnetism and Matter

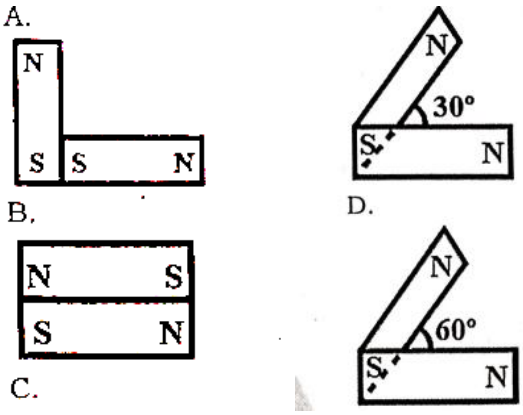
1. Magnetic field. The net magnetic force on the dipole
1) Is always zero
2) Depends on the orientation of the dipole
3) Can never be zero
4) Depends on the strength of the dipole

2. A magnet of magnetic moment M and pole strength m is divided in two equal parts. Then magnetic moment of each part will be
1) M
2) M/2
3) M/4
4) 2M

3. Two lines of force due to a bar magnet
1) Intersect at the neutral point
2) Intersect near the poles of the magnet
3) Intersect on the equatorial axis of the magnet
4) Do not intersect at all

4. The torque acting on a bar magnet of magnetic moment Min a uniform magnetic field B will be
1) MB $\sin\theta$
2) MB/ $\sin\theta$
3) MB/ $\cos\theta$
4) MB/ θ

5. Following figures show the arrangement of bar magnets in different configurations. Each magnet has magnetic dipole moment m. Which configuration has highest net magnetic dipole moment.



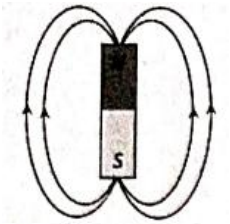
1) C
2) D
3) A
4) B

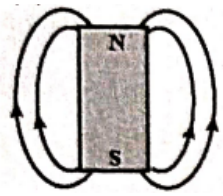
6. The magnetic moment of a magnet of length 10 cm and pole strength 4.0 Am will be
1) $0.4\,Am^2$
2) $1.6\,Am^2$
3) $20\,Am^2$
4) $8.0\,Am^2$

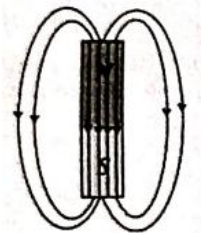
7. The magnetic lines of force inside a bar magnet

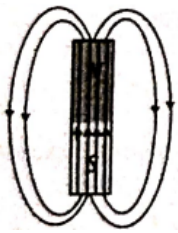
1) Are from south-pole to north-pole of the magnet
2) Are from north-pole to south-pole of the magnet
3) Do not exist
4) Depend upon the area of cross-section of the bar magnet

8. The magnetic field lines due to a bar magnet are correctly shown in

1) 

2) 

3) 

4) 

9. A short bar magnet of magnetic moment $0.4\,JT^{-1}$ is placed in a uniform magnetic field of 0.16 T. The magnet is in stable equilibrium when the potential energy is

1) $-0.082\,J$ 2) $0.064\,J$ 3) $-0.064\,J$ 4) Zero

10. If a bar magnet of magnetic moment M is freely suspended in a uniform magnetic field of strength B, the work done in rotating the magnet through an angle θ is

1) $MB(1-sin\theta)$
2) $MB\,sin\,\theta$
3) $MB\,cos\,\theta$
4) $MB(1-\cos\theta)$

11. A magnetic compass needle which is allowed to move in a horizontal plane is taken to a geomagnetic pole. It

1) Will become rigid showing no movement
2) Will stay in any position
3) Will stay in north-south direction only
4) Will stay in east-west direction only

12. The angle of dip a certain place is 30°. If the horizontal component of the earth's magnetic field is H, the intensity of the total magnetic field is

1) $\frac{2H}{\sqrt{3}}$
2) $\frac{2H}{\sqrt{3}}$
3) $H\sqrt{2}$
4) $H\sqrt{3}$

13. Magnetic meridian is a

1) Point
2) Horizontal plane
3) Vertical plane
4) Line along N-S

14. At a certain place, the horizontal component of earth's magnetic field is $\sqrt{3}$ times the vertical component. The angle of dip at that place is

1) 60^0
2) 45^0
3) 90^0
4) 30^0

15. If θ_1 and θ_2 be the apparent angles of dip observed in two vertical planes at right angles to each other, then the true angle of dip θ is given by

1) $\cot^2\theta = \cot^2\theta_1 + \cot^2\theta_2$
2) $\tan^2\theta = \tan^2\theta_1 + \tan^2\theta_2$
3) $\cot^2\theta = \cot^2\theta_1 - \cot^2\theta_2$
4) $\tan^2\theta = \tan^2\theta_1 - \tan^2\theta_2$

16. A relative permittivity and permeability of a material are ε_r and μ_r, respectively. Which of the following values of these quantities are allowed for a diamagnetic material

1) $\varepsilon_r = 1, \mu_r = 0.5$
2) $\varepsilon_r = 0.5, \mu_r = 0.5$
3) $\varepsilon_r = 1.5, \mu_r = 0.5$
4) $\varepsilon_r = 0.5, \mu_r = 1.5$

17. For diamagnetic materials, magnetic susceptibility is

1) Small and negative
2) Small and positive
3) Large and negative
4) Large and positive

18. Core of electromagnets are made of ferromagnetic material which has

1) High permeability and high retentivity
2) Low permeability and low retentivity
3) High permeability and low retentivity
4) Low permeability and high retentivity

19. The material of permanent magnet has

1) High retentivity, low coercivity
2) Low retentivity, high coercivity
3) Low retentivity, low coercivity
4) High retentivity, high coercivity

20. A paramagnetic substance of susceptibility 3×10^{-4} is placed in a magnetic field of $4\times10^{-7}\,Am^{-1}$. Then the intensity of magnetization in the units of Am^{-1} is

1) 1.33×10^{-8}
2) 0.75×10^{8}
3) 12×10^{-8}
4) 14×10^{8}

21. The only property possessed by ferromagnetic substance is

1) Hysteresis
2) Susceptibility
3) Directional property

4) Attracting magnetic substances

22. Which one of the following is not a characteristics of diamagnetism

1) The diamagnetic materials are repelled by a bar magnet.
2) The magnetic susceptibility of the materials are small and negative.
3) The origin of diamagnetism is the spin of electros
4) The material moves from a region of strong magnetic field to weak magnetic field

23. Relative permeability of iron is 5500, then its magnetic susceptibility will be

1) 5000×10^{7}
2) 5000×10^{-7}
3) 5501
4) 5499

24. The magnetic susceptibility of a paramagnetic substance at -73°C is 0.0075, then its value at –173°C will be

1) 0.0030
2) 0.0150
3) 0.0180
4) 0.0045

25. Susceptibility of ferromagnetic substance is

1) >1
2) <1
3) 0
4) 1

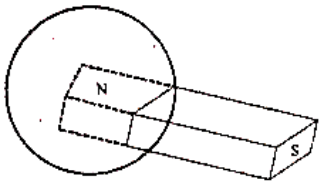
26. Statement A: The poles of magnet cannot be separated by breaking the magnet into two pieces.
Statement B: Magnetic monopoles do not exist.

1) Statement A is correct
2) Statement B is correct
3) Both A and B are correct
4) Both A and B are incorrect

27. Which of the following statements is correct?

1) Magnetic flux is a vector quantity.
2) Two magnetic field lines may intersect
3) The net magnetic flux through any closed surface is zero
4) Earth's magnetic poles coincide with the geographic poles.

28. A spherical surface of radius 'r' is enclosing the north pole of a bar magnet as shown below. The magnetic flux through the spherical surface is.



1) Infinity
2) $\frac{\mu_0}{4\pi r}$
3) $2\pi\mu_0 r$
4) Zero

29. Which of the following option gives correct equation for Gauss' Law of magnetism?

1) $\oint \vec{E}.\overrightarrow{dA} = \frac{q}{\epsilon_0}$

2) $\oint \vec{E}.\overrightarrow{dA} = 0$

3) $\oint \vec{B}.\overrightarrow{dA} = 0$

4) None of these

30. A magnetic field of 8.9 T passes perpendicular to a disc with a radius of 5 cm. Find the magnetic flux of disc

1) $6.9 \times 10^{-2} Wb$

2) $3.45 \times 10^{-2} Wb$

3) $3.45 \times 10^{-3} Wb$

4) $1.34 \times 10^{-2} Wb$

Answer Key

(1)	**1**	(16)	**3**
(2)	**2**	(17)	**1**
(3)	**4**	(18)	**3**
(4)	**1**	(19)	**4**
(5)	**1**	(20)	**3**
(6)	**1**	(21)	**1**
(7)	**1**	(22)	**3**
(8)	**4**	(23)	**4**
(9)	**3**	(24)	**2**
(10)	**4**	(25)	**1**
(11)	**2**	(26)	**3**
(12)	**2**	(27)	**3**
(13)	**3**	(28)	**4**
(14)	**4**	(29)	**3**
(15)	**1**	(30)	**1**